

Stretching vs. Bending

What is the difference between bending and stretching?

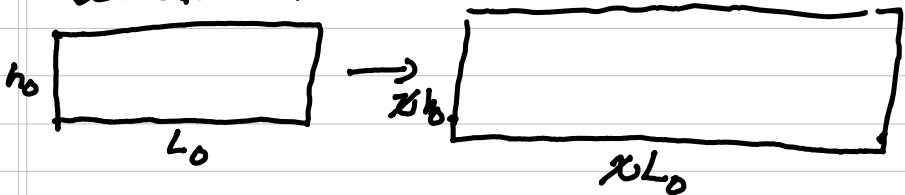
Imagine the following thing



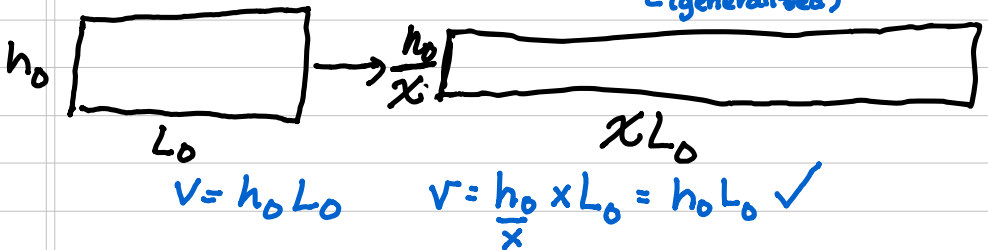
Stretching:

we expand it uniformly along one or more axes.

isotropic: preserve aspect ratio



volume conserving: preserve "volume"
 $\hookrightarrow$ (generalized)



We are not changing the shape.



For membranes most stretching is "volume" conserving:

area $\downarrow$, thickness $\uparrow$
or

area $\uparrow$, thickness $\downarrow$

Let's Taylor expand to get the stretching free energy:

$$f_s = f_1(\Sigma - \Sigma_0) + \frac{1}{2} f_2 (\Sigma - \Sigma_0)^2 + o(\Sigma - \Sigma_0)^3$$

Σ_0 = equilibrium area/molecule

$$f_1 = \left(\frac{\partial f}{\partial \Sigma} \right) \bigg|_{\Sigma = \Sigma_0} = 0 \text{ bc } \Sigma_0 \text{ is the equil area}$$

$$f_2 = \left(\frac{\partial^2 f}{\partial \Sigma^2} \right) \bigg|_{\Sigma_0}$$

$$f_s = \frac{1}{2} f_2 (\Sigma - \Sigma_0)^2 + o(\Sigma - \Sigma_0)^3$$

$\uparrow$ usually f_2 is written k_A / Σ_0 so...

$$f_s = \frac{1}{2} k_A \frac{(\Sigma - \Sigma_0)^2}{\Sigma_0} \quad \text{where } k_A \text{ is known as the "compressibility modulus"}$$

Previously:

$$f_s = \frac{1}{2} k_A \frac{(\Sigma - \Sigma_0)^2}{\Sigma_0}$$

The definition of the surface tension γ is

$$\gamma = \left(\frac{\partial F}{\partial \Sigma} \right)_V$$

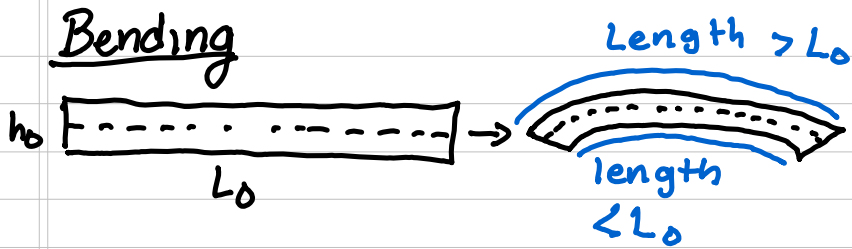
$$\frac{\partial f_s}{\partial \Sigma} = \frac{k_A (\Sigma - \Sigma_0)}{\Sigma_0} = \gamma$$

so you can write it like this

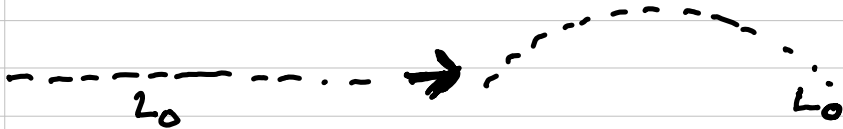
$$f_s = \frac{1}{2} k_A (\Sigma - \Sigma_0)^2 / \Sigma_0$$

$$f_s = \frac{1}{2} \overset{\text{or}}{\gamma} (\Sigma - \Sigma_0)^2$$

k_A is analogous to spring constant
 γ is analogous to restoring force



The dashed line represents the "neutral surface". It has length L_0 both before and after the transformation



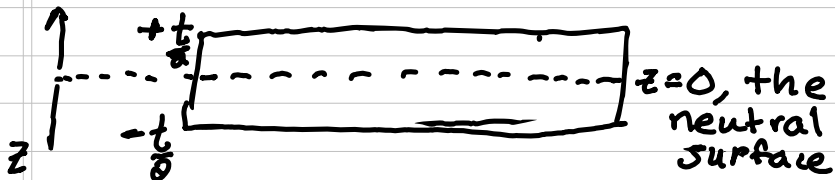
So, since the length hasn't changed, why should this deformation cost any energy?? It's true the average length hasn't changed, but most parts of the surface aren't at the average!!!



Bending is just a bit of stretching/compression

We can write membranes as a stack of infinitely thin sheets, each with their own stretching energy

$$f_s(z) = \frac{k_A}{2t} \frac{(\Sigma(z) - \Sigma_0)^2}{\Sigma_0}$$



To get the free energy of a bent surface, integrate over the stretching energy of all the sheets

$$f_c(H) = \frac{k_A}{2t} \int dz \frac{(\Sigma(z, H) - \Sigma_0)^2}{\Sigma_0}$$

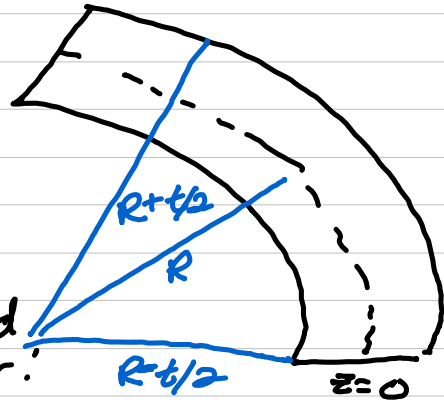
Exercise:

You take a flat sheet of thickness t and area Σ_0 , and bend it into a cylinder of radius R . The sheet has compressibility k_A . What is the free energy of this deformation? Assume the sheet is uniform and each horizontal slice has compressibility k_A/t .

Warning: this problem gives a nice answer but assumes a homogeneous sheet.

Solution

The radius of each slice is $R + z$. The ratio of areas of each slice is the same as if they included a whole cylinder.



$$\frac{\Sigma'(z)}{\Sigma(0)} = \frac{2\pi(R+z)\sqrt{\Sigma_0}}{2\pi R\sqrt{\Sigma_0}} = \frac{R+z}{R} = \frac{\delta'(z)}{\delta_0}$$

$$\Sigma(z) = \Sigma_0 \left(1 + \frac{z}{R} \right) \rightarrow \Sigma - \Sigma_0 = \frac{z}{R} \Sigma_0$$

So if we integrate over all the slices:

$$\frac{K_A \Sigma_0^2}{2 \Sigma_0 t} \int_{-t/2}^{t/2} dz \left(\frac{z}{R} \right)^2 = \frac{K_A \Sigma_0}{2 t R^2} \cdot \frac{1}{3} \left[\left(\frac{t}{2} \right)^3 - \left(\frac{-t}{2} \right)^3 \right]$$

$$= \frac{K_A \Sigma_0 t^2}{24 R^2} = \frac{K_A \Sigma_0 t^2 H^2}{6}$$

compare to curvature

compare to curvature free energy

$$2k_c H^2 \int dA = 2k_c H^2 \sum_0^x = \frac{k_A}{6} \sum_0^x t^2 H^2 \rightarrow \boxed{k_c = \frac{k_A t^2}{12}}$$